

ULTIMATE ERP

Developer Technical Documentation

Database Schema (ERD) & Backend API Interface Manual

Target Audience: Backend Engineers /
Integration Developers

Database Engine: MySQL / MariaDB
(InnoDB)

API Format: JSON-over-HTTP
(POST Action Router)

**Document
Version:** 1.0.0 (June 2026)

Table of Contents

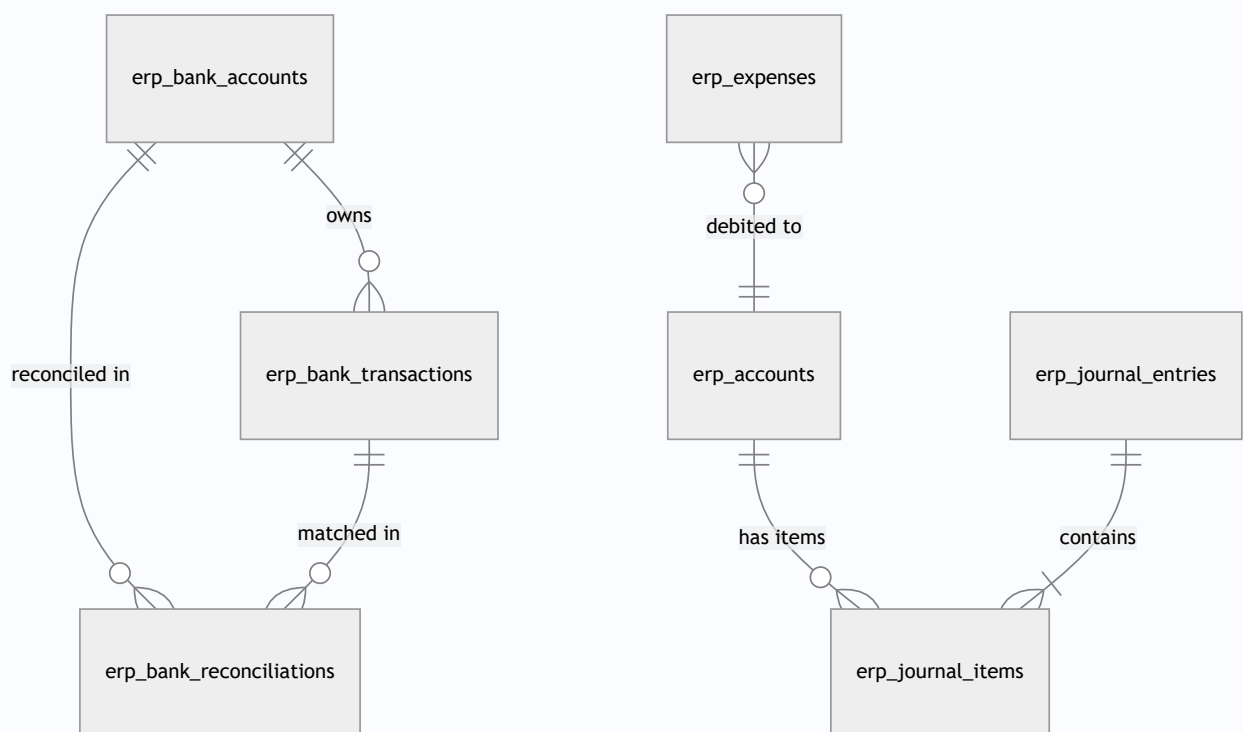
1. Database Architecture & ER Diagrams	Page 3
1.1 Core Finance & Ledger Module	Page 3
1.2 Sales & Receivables (AR) Module	Page 4
1.3 Purchases & Payables (AP) Module	Page 5
1.4 Stock & Inventory Module	Page 6
1.5 HR & Administration Module	Page 7
1.6 Users & Access Controls	Page 8
2. Database Table Descriptions	Page 9
3. Backend API Interface Manual	Page 18
3.1 API Overview & Protocol specifications	Page 18
3.2 API Reference Index	Page 19

1. Database Architecture & ER Diagrams

The database uses an InnoDB storage engine configured with UTF-8 character encoding to support transactions, row-level locking, and foreign key integrity. The system consists of 50 tables organized logically into 6 ERP modules. To maximize clarity, the entity-relationship diagrams have been decomposed by functional business area.

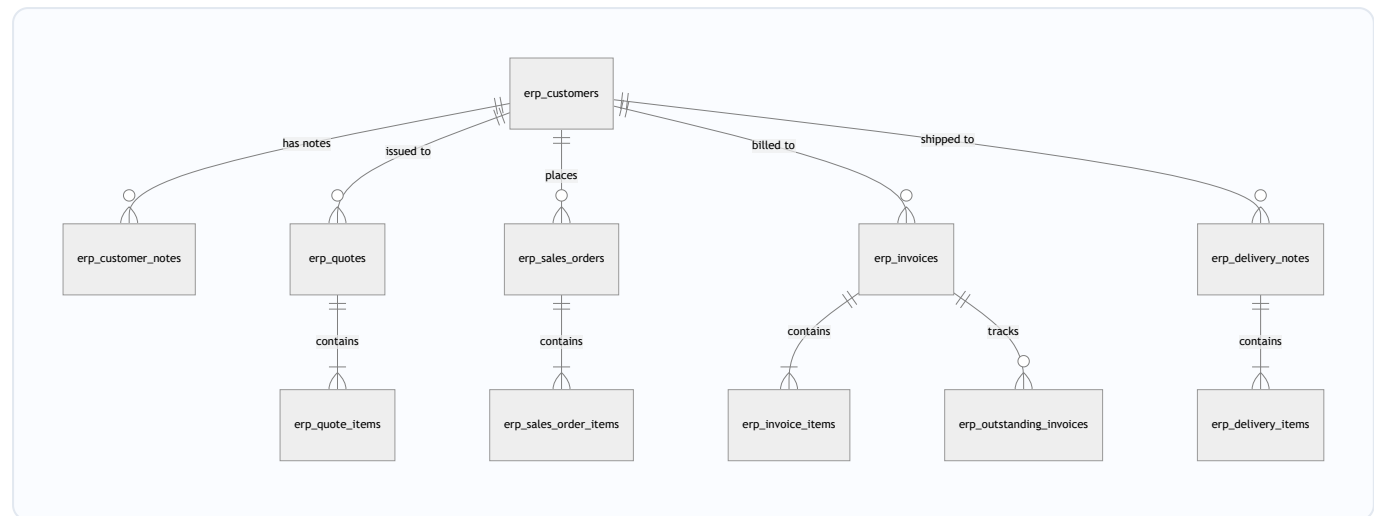
1.1 Core Finance & Ledger Module

This module houses the Chart of Accounts, General Ledger, journal postings, bank accounts, bank statement transaction feeds, bank reconciliation logs, tax settings, and general system configuration settings.



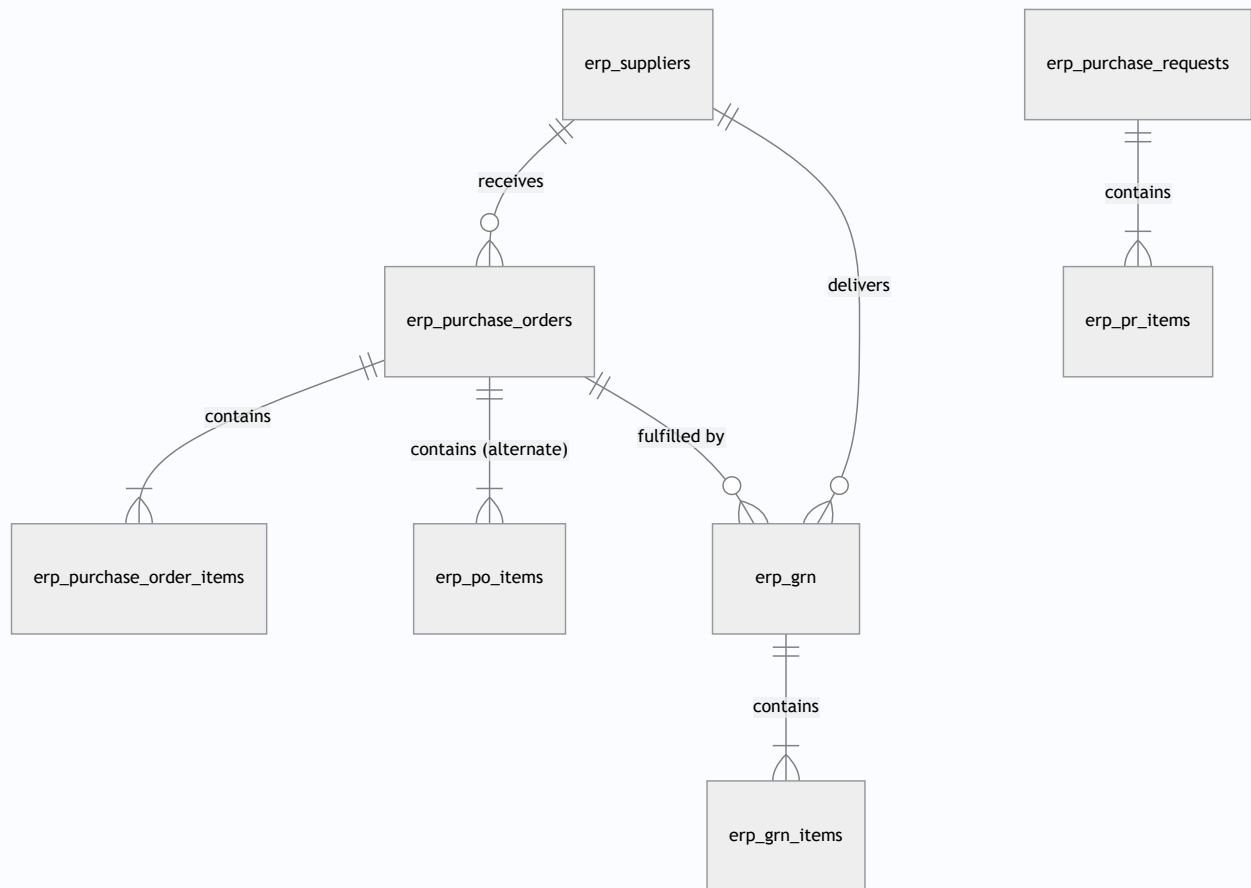
1.2 Sales & Receivables (AR) Module

Handles the customer master registry, customer statements and notes, quotes, sales orders, accounts receivable invoices, credit sales tracking, delivery notes, and logistics shipments.



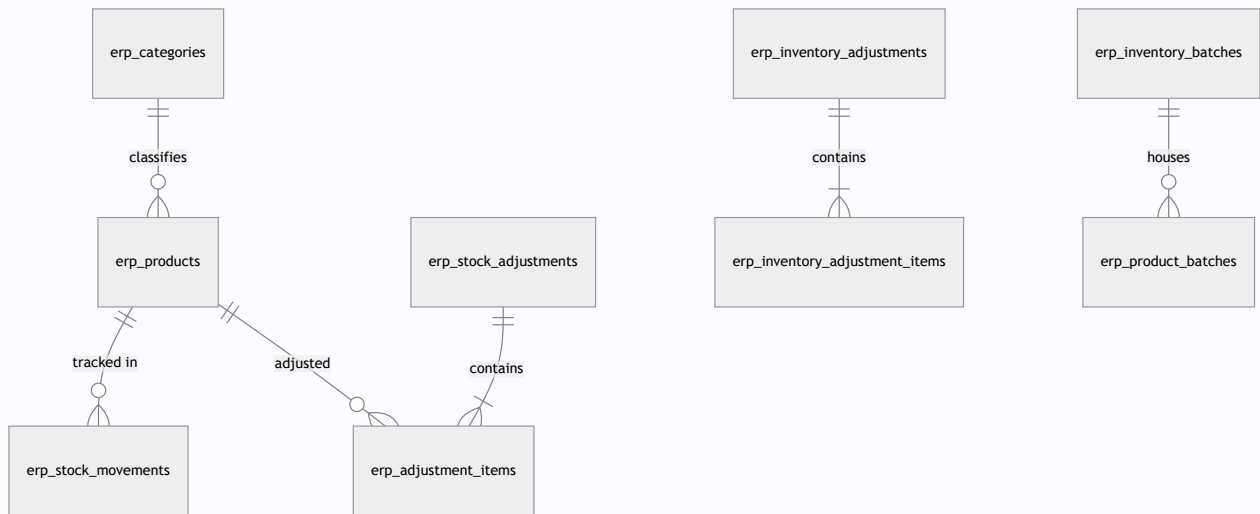
1.3 Purchases & Payables (AP) Module

Tracks suppliers, purchase requests (PR), purchase orders (PO), goods received notes (GRN), and linked detail items matching items received from vendors.



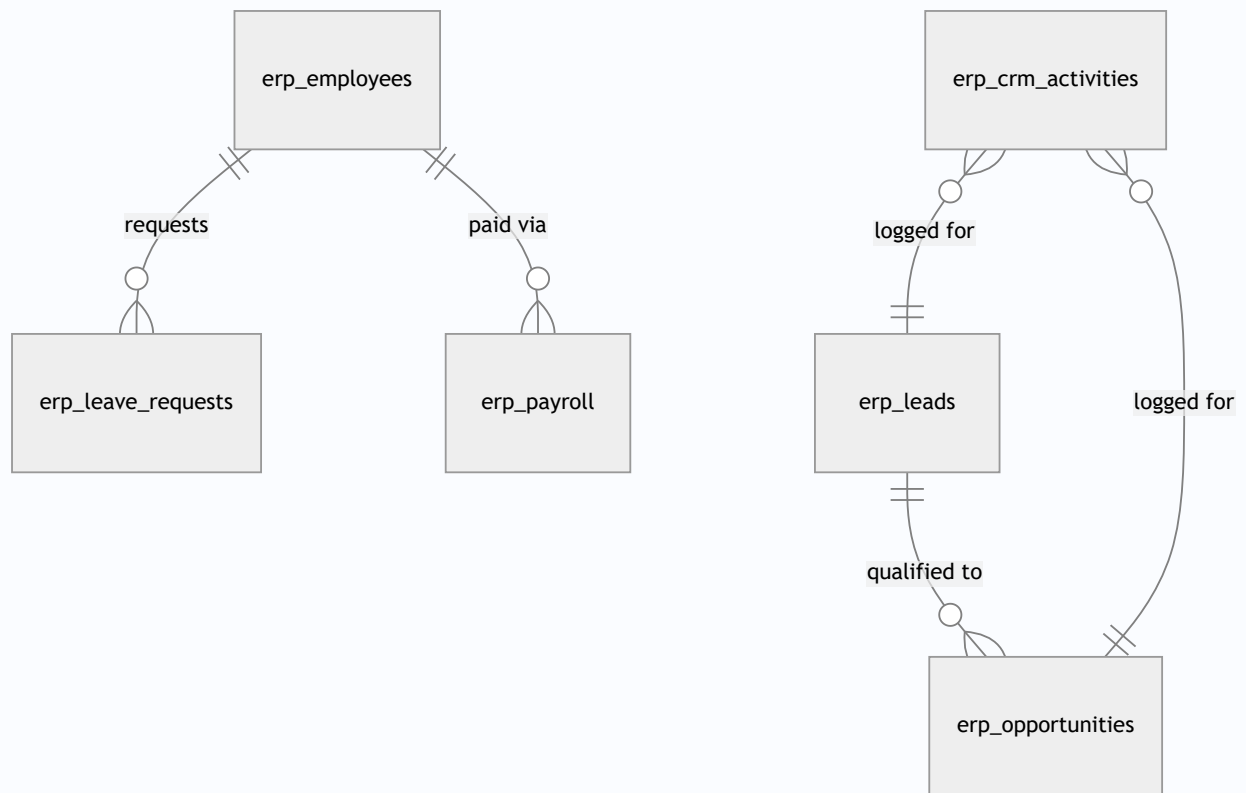
1.4 Stock & Inventory Module

Manages categories, product master lists, warehouses/locations, stock ledger entries (movements), physical stock adjustment logs, inventory batches, and expiry date schedules.



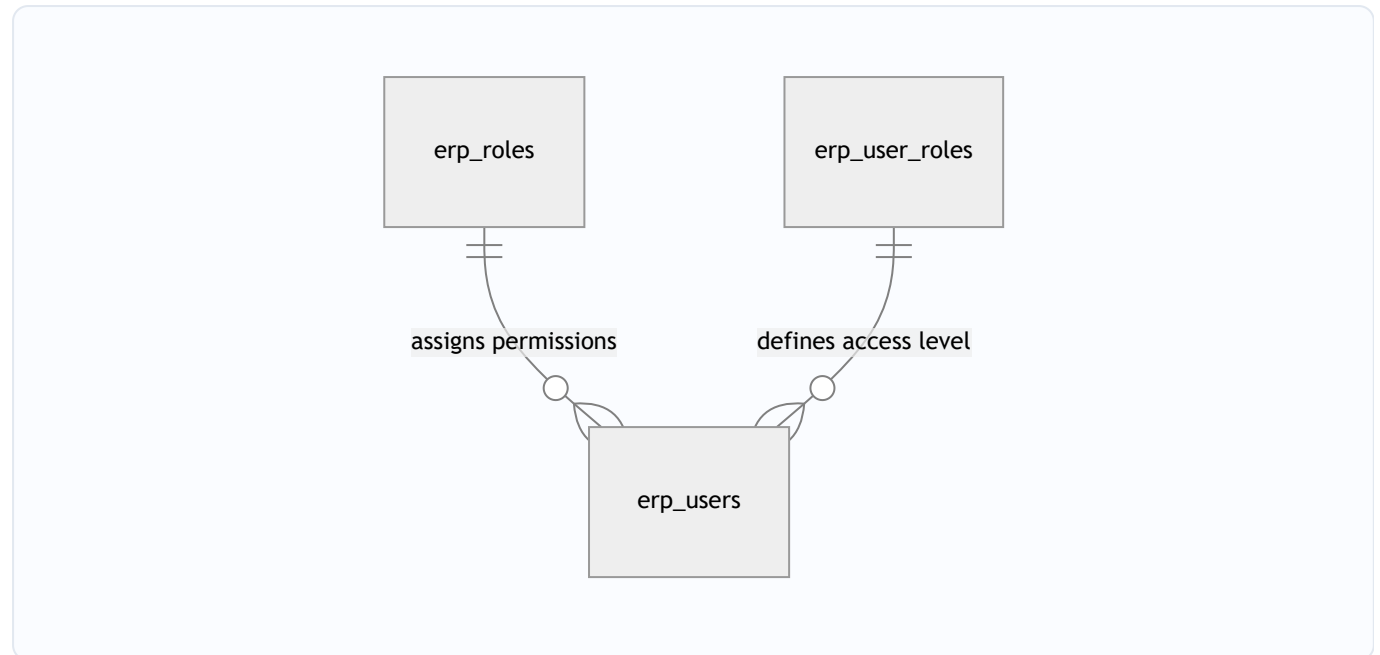
1.5 HR & Administration Module

Provides core employee databases, salary calculations (payroll runs), leave tracking, system activities logging, communications (activities, emails), notifications, and CRM (leads, deals, opportunities).



1.6 Users & Access Controls

Deals with user login credentials, tenant companies, branches, role definitions, and access permissions.



2. Database Table Descriptions

The following tables represent the detailed structures, fields, data types, and annotations of each entity in the database.

TABLE: `erp_accounts`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>code</code>	<code>varchar (255)</code>	Unique identification code
<code>name</code>	<code>varchar (255)</code>	Full name or name of the entity
<code>type</code>	<code>varchar (255)</code>	Classification type
<code>description</code>	<code>varchar (255)</code>	Detailed description or notes
<code>parent_id [FK]</code>	<code>int (11)</code>	Self-referencing parent ID for hierarchical structures
<code>is_system</code>	<code>varchar (255)</code>	Field: <code>is_system</code>
<code>status</code>	<code>varchar (255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

TABLE: `erp_activities`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>entity_type</code>	<code>varchar (50)</code>	Field: <code>entity_type</code>
<code>entity_id [FK]</code>	<code>int (11)</code>	Field: <code>entity_id</code>
<code>user_id [FK]</code>	<code>int (11)</code>	Reference to <code>erp_users(id)</code>
<code>action</code>	<code>varchar (50)</code>	Field: <code>action</code>
<code>description</code>	<code>text</code>	Detailed description or notes
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created

TABLE: `erp_adjustment_items`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>adjustment_id [FK]</code>	<code>int (11)</code>	Field: <code>adjustment_id</code>
<code>product_id [FK]</code>	<code>int (11)</code>	Reference to <code>erp_products(id)</code>
<code>old_quantity</code>	<code>decimal (15</code>	Field: <code>old_quantity</code>
<code>new_quantity</code>	<code>decimal (15</code>	Field: <code>new_quantity</code>
<code>difference</code>	<code>decimal (15</code>	Field: <code>difference</code>

TABLE: `erp_bank_accounts`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>account_name</code>	<code>varchar (255)</code>	Field: <code>account_name</code>
<code>account_number</code>	<code>varchar (255)</code>	Unique reference number
<code>bank_name</code>	<code>varchar (255)</code>	Field: <code>bank_name</code>
<code>branch</code>	<code>varchar (255)</code>	Field: <code>branch</code>
<code>currency</code>	<code>varchar (255)</code>	Field: <code>currency</code>
<code>opening_balance</code>	<code>decimal (10</code>	Field: <code>opening_balance</code>
<code>current_balance</code>	<code>decimal (10</code>	Field: <code>current_balance</code>
<code>gl_account_id [FK]</code>	<code>int (11)</code>	Field: <code>gl_account_id</code>
<code>status</code>	<code>varchar (255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

TABLE: `erp_bank_reconciliations`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>bank_account_id [FK]</code>	<code>int (11)</code>	Field: <code>bank_account_id</code>
<code>reconciliation_date</code>	<code>date</code>	Associated date timestamp
<code>statement_balance</code>	<code>decimal (15</code>	Field: <code>statement_balance</code>
<code>book_balance</code>	<code>decimal (15</code>	Field: <code>book_balance</code>
<code>difference</code>	<code>decimal (15</code>	Field: <code>difference</code>
<code>reconciled_by</code>	<code>int (11)</code>	Field: <code>reconciled_by</code>
<code>notes</code>	<code>text</code>	Internal remarks or customer-facing notes
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created

TABLE: `erp_bank_transactions`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>bank_account_id [FK]</code>	<code>int (11)</code>	Field: <code>bank_account_id</code>
<code>transaction_date</code>	<code>date</code>	Associated date timestamp
<code>description</code>	<code>varchar (255)</code>	Detailed description or notes
<code>reference</code>	<code>varchar (255)</code>	Field: <code>reference</code>
<code>debit</code>	<code>decimal (10</code>	Field: <code>debit</code>
<code>credit</code>	<code>decimal (10</code>	Field: <code>credit</code>
<code>balance</code>	<code>decimal (10</code>	Outstanding unpaid balance on the document
<code>reconciled</code>	<code>varchar (255)</code>	Field: <code>reconciled</code>
<code>created_by</code>	<code>int (11)</code>	User ID of the creator

TABLE: `erp_categories`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>name</code>	<code>varchar(255)</code>	Full name or name of the entity
<code>description</code>	<code>varchar(255)</code>	Detailed description or notes
<code>status</code>	<code>varchar(255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

TABLE: `erp_crm_activities`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>type</code>	<code>varchar(255)</code>	Classification type
<code>subject</code>	<code>varchar(255)</code>	Topic or header of the record
<code>description</code>	<code>varchar(255)</code>	Detailed description or notes
<code>lead_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_leads(id)</code>
<code>opportunity_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_opportunities(id)</code>
<code>customer_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_customers(id)</code>
<code>due_date</code>	<code>varchar(255)</code>	Target completion date
<code>status</code>	<code>varchar(255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>created_by</code>	<code>int(11)</code>	User ID of the creator

TABLE: `erp_customer_notes`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>customer_id [FK]</code>	<code>int (11)</code>	Reference to <code>erp_customers(id)</code>
<code>note</code>	<code>text</code>	Field: note
<code>created_by</code>	<code>int (11)</code>	User ID of the creator
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created

TABLE: `erp_customers`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>customer_code</code>	<code>varchar (255)</code>	Unique alphanumeric business code for the customer
<code>name</code>	<code>varchar (255)</code>	Full name or name of the entity
<code>email</code>	<code>varchar (255)</code>	Email address
<code>phone</code>	<code>varchar (255)</code>	Phone number
<code>address</code>	<code>varchar (255)</code>	Street/mailling address
<code>city</code>	<code>varchar (255)</code>	City
<code>country</code>	<code>varchar (255)</code>	Country (default: Tanzania)
<code>tax_id [FK]</code>	<code>varchar (255)</code>	TIN or VRN tax identification number
<code>credit_limit</code>	<code>decimal (10</code>	Maximum credit limit allowed for credit sales
<code>notes</code>	<code>varchar (255)</code>	Internal remarks or customer-facing notes
<code>status</code>	<code>varchar (255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>created_by</code>	<code>int (11)</code>	User ID of the creator
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created

TABLE: `erp_delivery_items`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>delivery_id [FK]</code>	<code>int(11)</code>	Field: <code>delivery_id</code>
<code>product_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_products(id)</code>
<code>quantity</code>	<code>decimal(10</code>	Line item quantity
<code>batch_number</code>	<code>varchar(255)</code>	Unique reference number

TABLE: `erp_delivery_notes`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>order_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_orders(id)</code>
<code>delivery_number</code>	<code>varchar(50)</code>	Unique reference number
<code>invoice_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_invoices(id)</code>
<code>customer_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_customers(id)</code>
<code>date</code>	<code>date</code>	Document date
<code>delivery_date</code>	<code>date</code>	Associated date timestamp
<code>status</code>	<code>varchar(255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>shipping_address</code>	<code>varchar(255)</code>	Field: <code>shipping_address</code>
<code>driver_name</code>	<code>varchar(255)</code>	Field: <code>driver_name</code>
<code>vehicle_number</code>	<code>varchar(255)</code>	Unique reference number
<code>vehicle_reg</code>	<code>varchar(50)</code>	Field: <code>vehicle_reg</code>
<code>notes</code>	<code>varchar(255)</code>	Internal remarks or customer-facing notes
<code>created_by</code>	<code>int(11)</code>	User ID of the creator

TABLE: `erp_departments`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>name</code>	<code>varchar(100)</code>	Full name or name of the entity
<code>description</code>	<code>text</code>	Detailed description or notes
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created

TABLE: `erp_email_templates`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>template_name</code>	<code>varchar(100)</code>	Field: <code>template_name</code>
<code>subject</code>	<code>varchar(255)</code>	Topic or header of the record
<code>body</code>	<code>text</code>	Field: <code>body</code>
<code>variables</code>	<code>text</code>	Field: <code>variables</code>
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created
<code>updated_at</code>	<code>timestamp</code>	Timestamp when the record was last updated

TABLE: `erp_employees`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>employee_code</code>	<code>varchar(255)</code>	Unique identification code
<code>first_name</code>	<code>varchar(255)</code>	Field: <code>first_name</code>
<code>last_name</code>	<code>varchar(255)</code>	Field: <code>last_name</code>
<code>email</code>	<code>varchar(255)</code>	Email address
<code>phone</code>	<code>varchar(255)</code>	Phone number
<code>department_id [FK]</code>	<code>int(11)</code>	Field: <code>department_id</code>
<code>position</code>	<code>varchar(255)</code>	Field: <code>position</code>
<code>join_date</code>	<code>date</code>	Associated date timestamp
<code>basic_salary</code>	<code>decimal(10</code>	Field: <code>basic_salary</code>
<code>bank_name</code>	<code>varchar(255)</code>	Field: <code>bank_name</code>
<code>bank_account_number</code>	<code>varchar(255)</code>	Unique reference number
<code>status</code>	<code>varchar(255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>user_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_users(id)</code>

TABLE: `erp_expenses`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>expense_number</code>	<code>varchar(255)</code>	Unique reference number
<code>date</code>	<code>date</code>	Document date
<code>payee</code>	<code>varchar(255)</code>	Field: payee
<code>account_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_accounts(id)</code>
<code>amount</code>	<code>decimal(10</code>	Financial value or cost
<code>payment_method</code>	<code>varchar(255)</code>	Field: payment_method
<code>description</code>	<code>varchar(255)</code>	Detailed description or notes
<code>status</code>	<code>varchar(255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>created_by</code>	<code>int(11)</code>	User ID of the creator
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created

TABLE: `erp_grn`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>grn_number</code>	<code>varchar(50)</code>	Unique reference number
<code>po_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_purchase_orders(id)</code>
<code>supplier_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_suppliers(id)</code>
<code>received_date</code>	<code>date</code>	Associated date timestamp
<code>received_by</code>	<code>int(11)</code>	User ID of the receiver
<code>delivery_note_ref</code>	<code>varchar(100)</code>	Supplier delivery note reference number
<code>status</code>	<code>enum(</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>notes</code>	<code>text</code>	Internal remarks or customer-facing notes
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created

TABLE: `erp_grn_items`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>grn_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_grn(id)</code>
<code>po_item_id [FK]</code>	<code>int(11)</code>	Field: <code>po_item_id</code>
<code>product_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_products(id)</code>
<code>quantity_ordered</code>	<code>decimal(10</code>	Field: <code>quantity_ordered</code>
<code>quantity_received</code>	<code>decimal(10</code>	Field: <code>quantity_received</code>
<code>quantity_rejected</code>	<code>decimal(10</code>	Field: <code>quantity_rejected</code>
<code>rejection_reason</code>	<code>text</code>	Field: <code>rejection_reason</code>
<code>batch_number</code>	<code>varchar(100)</code>	Unique reference number
<code>expiry_date</code>	<code>date</code>	Expiration date

TABLE: `erp_inventory_adjustment_items`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>adjustment_id [FK]</code>	<code>int(11)</code>	Field: <code>adjustment_id</code>
<code>product_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_products(id)</code>
<code>quantity_before</code>	<code>decimal(10</code>	Field: <code>quantity_before</code>
<code>quantity_after</code>	<code>decimal(10</code>	Field: <code>quantity_after</code>
<code>difference</code>	<code>decimal(10</code>	Field: <code>difference</code>

TABLE: `erp_inventory_adjustments`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>adjustment_number</code>	<code>varchar(255)</code>	Unique inventory adjustment document number
<code>date</code>	<code>date</code>	Document date
<code>reason</code>	<code>varchar(255)</code>	Explanation or justification
<code>notes</code>	<code>varchar(255)</code>	Internal remarks or customer-facing notes
<code>status</code>	<code>varchar(255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>created_by</code>	<code>int(11)</code>	User ID of the creator

TABLE: `erp_inventory_batches`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>product_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_products(id)</code>
<code>batch_number</code>	<code>varchar(255)</code>	Unique reference number
<code>expiry_date</code>	<code>date</code>	Expiration date
<code>quantity</code>	<code>decimal(10</code>	Line item quantity
<code>cost_price</code>	<code>decimal(10</code>	Standard purchase cost per unit

TABLE: `erp_invoice_items`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>invoice_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_invoices(id)</code>
<code>product_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_products(id)</code>
<code>description</code>	<code>varchar(255)</code>	Detailed description or notes
<code>quantity</code>	<code>decimal(10</code>	Line item quantity
<code>unit_price</code>	<code>decimal(10</code>	Standard selling price per unit
<code>tax_rate</code>	<code>decimal(10</code>	Applied VAT percentage rate (e.g., 18.00)
<code>total</code>	<code>decimal(10</code>	Grand total amount (subtotal + tax - discount)

TABLE: `erp_invoices`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>invoice_number</code>	<code>varchar (255)</code>	Unique system invoice number
<code>customer_id [FK]</code>	<code>int (11)</code>	Reference to <code>erp_customers(id)</code>
<code>invoice_date</code>	<code>date</code>	Date the invoice was issued
<code>due_date</code>	<code>date</code>	Target completion date
<code>subtotal</code>	<code>decimal (10</code>	Total amount before taxes and discounts
<code>tax_rate</code>	<code>decimal (10</code>	Applied VAT percentage rate (e.g., 18.00)
<code>tax_amount</code>	<code>decimal (10</code>	Tax amount for the specific line or document
<code>total</code>	<code>decimal (10</code>	Grand total amount (subtotal + tax - discount)
<code>balance</code>	<code>decimal (10</code>	Outstanding unpaid balance on the document
<code>notes</code>	<code>varchar (255)</code>	Internal remarks or customer-facing notes
<code>status</code>	<code>varchar (255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>created_by</code>	<code>int (11)</code>	User ID of the creator
<code>created_at</code>	<code>varchar (255)</code>	Timestamp when the record was created

TABLE: `erp_journal_entries`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>entry_number</code>	<code>varchar(255)</code>	Unique reference number
<code>date</code>	<code>date</code>	Document date
<code>description</code>	<code>varchar(255)</code>	Detailed description or notes
<code>status</code>	<code>varchar(255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>created_by</code>	<code>int(11)</code>	User ID of the creator

TABLE: `erp_journal_items`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>journal_id [FK]</code>	<code>int(11)</code>	Field: <code>journal_id</code>
<code>account_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_accounts(id)</code>
<code>debit</code>	<code>decimal(10)</code>	Field: <code>debit</code>
<code>credit</code>	<code>decimal(10)</code>	Field: <code>credit</code>

TABLE: `erp_leads`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>first_name</code>	<code>varchar (255)</code>	Field: first_name
<code>last_name</code>	<code>varchar (255)</code>	Field: last_name
<code>email</code>	<code>varchar (255)</code>	Email address
<code>phone</code>	<code>varchar (255)</code>	Phone number
<code>company</code>	<code>varchar (255)</code>	Field: company
<code>source</code>	<code>varchar (255)</code>	Field: source
<code>status</code>	<code>varchar (255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>assigned_to</code>	<code>int (11)</code>	User ID of the assignee
<code>notes</code>	<code>varchar (255)</code>	Internal remarks or customer-facing notes

TABLE: `erp_leave_requests`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>employee_id [FK]</code>	<code>int (11)</code>	Reference to <code>erp_employees(id)</code>
<code>leave_type</code>	<code>varchar (255)</code>	Type of leave (e.g., 'sick', 'annual')
<code>start_date</code>	<code>date</code>	Leave starting date
<code>end_date</code>	<code>date</code>	Leave ending date
<code>reason</code>	<code>varchar (255)</code>	Explanation or justification
<code>status</code>	<code>varchar (255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>approved_by</code>	<code>int (11)</code>	Field: approved_by

TABLE: erp_notifications

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>user_id [FK]</code>	<code>int (11)</code>	Reference to <code>erp_users(id)</code>
<code>title</code>	<code>varchar (255)</code>	Field: title
<code>message</code>	<code>varchar (255)</code>	Message text content
<code>type</code>	<code>varchar (255)</code>	Classification type
<code>link</code>	<code>varchar (255)</code>	Field: link
<code>is_read</code>	<code>varchar (255)</code>	Field: is_read
<code>created_at</code>	<code>varchar (255)</code>	Timestamp when the record was created

TABLE: erp_opportunities

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>name</code>	<code>varchar (255)</code>	Full name or name of the entity
<code>customer_id [FK]</code>	<code>int (11)</code>	Reference to <code>erp_customers(id)</code>
<code>lead_id [FK]</code>	<code>int (11)</code>	Reference to <code>erp_leads(id)</code>
<code>amount</code>	<code>decimal (10)</code>	Financial value or cost
<code>stage</code>	<code>varchar (255)</code>	Sales pipeline stage
<code>probability</code>	<code>int (11)</code>	Estimated win probability percentage
<code>expected_close_date</code>	<code>date</code>	Target close date of opportunity
<code>assigned_to</code>	<code>int (11)</code>	User ID of the assignee
<code>notes</code>	<code>varchar (255)</code>	Internal remarks or customer-facing notes

TABLE: `erp_outstanding_invoices`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>type</code>	<code>enum(</code>	Classification type
<code>invoice_date</code>	<code>date</code>	Date the invoice was issued
<code>entity_name</code>	<code>varchar(255)</code>	Field: <code>entity_name</code>
<code>invoice_number</code>	<code>varchar(50)</code>	Unique system invoice number
<code>narration</code>	<code>text</code>	Field: <code>narration</code>
<code>amount</code>	<code>decimal(15</code>	Financial value or cost
<code>attachment</code>	<code>varchar(255)</code>	Field: <code>attachment</code>
<code>status</code>	<code>enum(</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created
<code>updated_at</code>	<code>timestamp</code>	Timestamp when the record was last updated

TABLE: erp_payroll

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>payroll_month</code>	<code>varchar(255)</code>	Field: payroll_month
<code>employee_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_employees(id)</code>
<code>basic_salary</code>	<code>decimal(10</code>	Field: basic_salary
<code>allowances</code>	<code>decimal(10</code>	Field: allowances
<code>deductions</code>	<code>decimal(10</code>	Field: deductions
<code>net_salary</code>	<code>decimal(10</code>	Field: net_salary
<code>status</code>	<code>varchar(255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>created_by</code>	<code>int(11)</code>	User ID of the creator

TABLE: erp_po_items

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>po_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_purchase_orders(id)</code>
<code>product_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_products(id)</code>
<code>description</code>	<code>varchar(500)</code>	Detailed description or notes
<code>quantity</code>	<code>decimal(15</code>	Line item quantity
<code>unit_price</code>	<code>decimal(15</code>	Standard selling price per unit
<code>total</code>	<code>decimal(15</code>	Grand total amount (subtotal + tax - discount)

TABLE: `erp_pr_items`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>pr_id [FK]</code>	<code>int (11)</code>	Reference to <code>erp_purchase_requests(id)</code>
<code>product_name</code>	<code>varchar (255)</code>	Field: <code>product_name</code>
<code>quantity</code>	<code>decimal (10</code>	Line item quantity
<code>unit</code>	<code>varchar (50)</code>	Unit of measure (e.g., 'pcs', 'kg')
<code>estimated_unit_cost</code>	<code>decimal (15</code>	Field: <code>estimated_unit_cost</code>
<code>total_cost</code>	<code>decimal (15</code>	Field: <code>total_cost</code>
<code>product_id [FK]</code>	<code>int (11)</code>	Reference to <code>erp_products(id)</code>

TABLE: `erp_product_batches`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>product_id [FK]</code>	<code>int (11)</code>	Reference to <code>erp_products(id)</code>
<code>batch_number</code>	<code>varchar (100)</code>	Unique reference number
<code>quantity</code>	<code>decimal (15</code>	Line item quantity
<code>remaining_quantity</code>	<code>decimal (10</code>	Field: <code>remaining_quantity</code>
<code>cost_price</code>	<code>decimal (15</code>	Standard purchase cost per unit
<code>received_date</code>	<code>date</code>	Associated date timestamp
<code>manufacturing_date</code>	<code>date</code>	Associated date timestamp
<code>expiry_date</code>	<code>date</code>	Expiration date
<code>status</code>	<code>enum (</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created

TABLE: `erp_products`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>sku</code>	<code>varchar(255)</code>	Stock Keeping Unit (unique product code)
<code>barcode</code>	<code>varchar(100)</code>	Universal product barcode number
<code>image_path</code>	<code>varchar(255)</code>	Field: <code>image_path</code>
<code>name</code>	<code>varchar(255)</code>	Full name or name of the entity
<code>description</code>	<code>varchar(255)</code>	Detailed description or notes
<code>category_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_categories(id)</code>
<code>supplier_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_suppliers(id)</code>
<code>unit</code>	<code>varchar(255)</code>	Unit of measure (e.g., 'pcs', 'kg')
<code>unit_price</code>	<code>decimal(10</code>	Standard selling price per unit
<code>cost_price</code>	<code>decimal(10</code>	Standard purchase cost per unit
<code>stock_quantity</code>	<code>decimal(10</code>	Current physical stock quantity on hand
<code>reorder_level</code>	<code>decimal(10</code>	Stock level threshold that triggers replenishment alerts
<code>status</code>	<code>varchar(255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>created_by</code>	<code>int(11)</code>	User ID of the creator
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created
<code>updated_at</code>	<code>timestamp</code>	Timestamp when the record was last updated

TABLE: `erp_purchase_order_items`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>po_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_purchase_orders(id)</code>
<code>product_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_products(id)</code>
<code>description</code>	<code>varchar(255)</code>	Detailed description or notes
<code>quantity</code>	<code>decimal(10</code>	Line item quantity
<code>unit_price</code>	<code>decimal(10</code>	Standard selling price per unit
<code>total</code>	<code>decimal(10</code>	Grand total amount (subtotal + tax - discount)
<code>received_quantity</code>	<code>decimal(10</code>	Field: <code>received_quantity</code>

TABLE: `erp_purchase_orders`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>po_number</code>	<code>varchar(255)</code>	Unique purchase order number
<code>supplier_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_suppliers(id)</code>
<code>order_date</code>	<code>date</code>	Purchase order date
<code>expected_delivery</code>	<code>date</code>	Field: <code>expected_delivery</code>
<code>subtotal</code>	<code>decimal(10</code>	Total amount before taxes and discounts
<code>tax_amount</code>	<code>decimal(10</code>	Tax amount for the specific line or document
<code>total</code>	<code>decimal(10</code>	Grand total amount (subtotal + tax - discount)
<code>notes</code>	<code>varchar(255)</code>	Internal remarks or customer-facing notes
<code>status</code>	<code>varchar(255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

TABLE: `erp_purchase_requests`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>request_number</code>	<code>varchar(50)</code>	Unique reference number
<code>request_date</code>	<code>date</code>	Associated date timestamp
<code>requested_by</code>	<code>int(11)</code>	Field: <code>requested_by</code>
<code>department</code>	<code>varchar(100)</code>	Field: <code>department</code>
<code>status</code>	<code>enum(</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>approval_level</code>	<code>int(11)</code>	Field: <code>approval_level</code>
<code>notes</code>	<code>text</code>	Internal remarks or customer-facing notes
<code>total_estimated_cost</code>	<code>decimal(15</code>	Field: <code>total_estimated_cost</code>
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created
<code>updated_at</code>	<code>timestamp</code>	Timestamp when the record was last updated

TABLE: `erp_quote_items`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>quote_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_quotes(id)</code>
<code>product_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_products(id)</code>
<code>description</code>	<code>varchar(255)</code>	Detailed description or notes
<code>quantity</code>	<code>decimal(10</code>	Line item quantity
<code>unit_price</code>	<code>decimal(10</code>	Standard selling price per unit
<code>tax_rate</code>	<code>decimal(10</code>	Applied VAT percentage rate (e.g., 18.00)
<code>total</code>	<code>decimal(10</code>	Grand total amount (subtotal + tax - discount)

TABLE: `erp_quotes`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>quote_number</code>	<code>varchar (255)</code>	Unique reference number
<code>customer_id [FK]</code>	<code>int (11)</code>	Reference to <code>erp_customers(id)</code>
<code>date</code>	<code>date</code>	Document date
<code>expiry_date</code>	<code>date</code>	Expiration date
<code>subtotal</code>	<code>decimal (10</code>	Total amount before taxes and discounts
<code>tax_amount</code>	<code>decimal (10</code>	Tax amount for the specific line or document
<code>total_amount</code>	<code>decimal (10</code>	Monetary value
<code>status</code>	<code>varchar (255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>notes</code>	<code>varchar (255)</code>	Internal remarks or customer-facing notes
<code>created_by</code>	<code>int (11)</code>	User ID of the creator
<code>updated_at</code>	<code>timestamp</code>	Timestamp when the record was last updated

TABLE: `erp_roles`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>company_id [FK]</code>	<code>int (11)</code>	Tenant ID (multi-company isolation)
<code>name</code>	<code>varchar (50)</code>	Full name or name of the entity
<code>permissions</code>	<code>longtext</code>	JSON/text representation of role permissions

TABLE: `erp_sales_order_items`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>order_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_orders(id)</code>
<code>product_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_products(id)</code>
<code>description</code>	<code>text</code>	Detailed description or notes
<code>quantity</code>	<code>decimal(10)</code>	Line item quantity
<code>unit_price</code>	<code>decimal(15)</code>	Standard selling price per unit
<code>tax_amount</code>	<code>decimal(15)</code>	Tax amount for the specific line or document
<code>total</code>	<code>decimal(15)</code>	Grand total amount (subtotal + tax - discount)

TABLE: `erp_sales_orders`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>order_number</code>	<code>varchar(50)</code>	Unique reference number
<code>quote_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_quotes(id)</code>
<code>customer_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_customers(id)</code>
<code>order_date</code>	<code>date</code>	Purchase order date
<code>delivery_date</code>	<code>date</code>	Associated date timestamp
<code>subtotal</code>	<code>decimal(15</code>	Total amount before taxes and discounts
<code>tax_amount</code>	<code>decimal(15</code>	Tax amount for the specific line or document
<code>total_amount</code>	<code>decimal(15</code>	Monetary value
<code>status</code>	<code>enum(</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>delivery_status</code>	<code>enum(</code>	Field: <code>delivery_status</code>
<code>invoice_status</code>	<code>enum(</code>	Field: <code>invoice_status</code>
<code>notes</code>	<code>text</code>	Internal remarks or customer-facing notes
<code>created_by</code>	<code>int(11)</code>	User ID of the creator
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created
<code>updated_at</code>	<code>timestamp</code>	Timestamp when the record was last updated

TABLE: `erp_settings`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>setting_key</code>	<code>varchar(255)</code>	Unique configuration parameter key
<code>setting_value</code>	<code>varchar(255)</code>	Configuration parameter value

TABLE: `erp_stock_adjustments`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>adjustment_number</code>	<code>varchar(50)</code>	Unique inventory adjustment document number
<code>adjustment_date</code>	<code>date</code>	Date of inventory adjustment
<code>reason</code>	<code>varchar(255)</code>	Explanation or justification
<code>notes</code>	<code>text</code>	Internal remarks or customer-facing notes
<code>created_by</code>	<code>int(11)</code>	User ID of the creator
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created

TABLE: `erp_stock_movements`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>product_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_products(id)</code>
<code>type</code>	<code>varchar(255)</code>	Classification type
<code>quantity</code>	<code>decimal(10</code>	Line item quantity
<code>reference_type</code>	<code>varchar(255)</code>	Source transaction type (e.g., 'Invoice', 'GRN')
<code>reference_id [FK]</code>	<code>int(11)</code>	Source transaction primary key
<code>notes</code>	<code>varchar(255)</code>	Internal remarks or customer-facing notes
<code>created_by</code>	<code>int(11)</code>	User ID of the creator
<code>created_at</code>	<code>varchar(255)</code>	Timestamp when the record was created

TABLE: `erp_suppliers`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>supplier_code</code>	<code>varchar (255)</code>	Unique alphanumeric business code for the supplier
<code>name</code>	<code>varchar (255)</code>	Full name or name of the entity
<code>email</code>	<code>varchar (255)</code>	Email address
<code>phone</code>	<code>varchar (255)</code>	Phone number
<code>address</code>	<code>varchar (255)</code>	Street/mailling address
<code>city</code>	<code>varchar (255)</code>	City
<code>country</code>	<code>varchar (255)</code>	Country (default: Tanzania)
<code>tax_id [FK]</code>	<code>varchar (255)</code>	TIN or VRN tax identification number
<code>notes</code>	<code>varchar (255)</code>	Internal remarks or customer-facing notes
<code>status</code>	<code>varchar (255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

TABLE: `erp_tax_rates`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int (11)</code>	Primary key, unique identifier (auto-incrementing)
<code>name</code>	<code>varchar (100)</code>	Full name or name of the entity
<code>rate</code>	<code>decimal (5</code>	Field: rate
<code>type</code>	<code>enum (</code>	Classification type
<code>is_default</code>	<code>tinyint (1)</code>	Boolean flag for default settings (1 = Yes, 0 = No)
<code>status</code>	<code>enum (</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created

TABLE: `erp_user_roles`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>role_name</code>	<code>varchar(100)</code>	Unique name of the system role
<code>description</code>	<code>text</code>	Detailed description or notes
<code>permissions</code>	<code>text</code>	JSON/text representation of role permissions
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created
<code>updated_at</code>	<code>timestamp</code>	Timestamp when the record was last updated
<code>status</code>	<code>varchar(255)</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

TABLE: `erp_users`

Field	Type	Description / Meanings
<code>id [PK]</code>	<code>int(11)</code>	Primary key, unique identifier (auto-incrementing)
<code>company_id [FK]</code>	<code>int(11)</code>	Tenant ID (multi-company isolation)
<code>branch_id [FK]</code>	<code>int(11)</code>	Reference to branch table
<code>email</code>	<code>varchar(100)</code>	Email address
<code>password_hash</code>	<code>varchar(255)</code>	Secure bcrypt hash of user password
<code>full_name</code>	<code>varchar(100)</code>	Full display name of the user
<code>role_id [FK]</code>	<code>int(11)</code>	Reference to <code>erp_roles(id)</code>
<code>is_active</code>	<code>tinyint(4)</code>	System access status (1 = Active, 0 = Inactive)
<code>created_at</code>	<code>timestamp</code>	Timestamp when the record was created

3. Backend API Interface Manual

3.1 API Overview & Protocol Specifications

The system backend APIs are structured as standard PHP endpoints residing in the `/api/` directory. Access follows these rules:

- **HTTP Request Method:** APIs are accessed via `POST` requests (with action-specific routing parameters passed in the request body).
- **Response Format:** All APIs return responses in `application/json` format.
- **General Response Structure:**

```
{
  "success": true | false,
  "message": "Status description or error message string",
  "id": 123 // Optional: returned on successful creations (lastInsertId)
}
```

- **Action Router Paradigm:** Endpoints utilize an `action` parameter passed in the POST body to route request execution to the proper controller logic.

3.2 API Reference Index

This index provides details for each API endpoint, the HTTP method, parameter inputs, and descriptions of operational actions.

/api/accounts.php

POST

Endpoint File: accounts.php

Available Routing Actions:

- list
- get_next_code
- create
- update
- delete

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
type	Classification type
name	Full name or name of the entity
code	Unique identification code
parent_id	Self-referencing parent ID for hierarchical structures
description	Detailed description or notes
id	Primary key, unique identifier (auto-incrementing)

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/adjustments.php

POST

Endpoint File: adjustments.php

Available Routing Actions:

- create

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
date	Document date
reason	Explanation or justification
items	Field: items
adjustment_number	Unique inventory adjustment document number
notes	Internal remarks or customer-facing notes

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/ai/ask.php

POST

Endpoint File: ai/ask.php

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

`/api/ai/settings.php`

POST

Endpoint File: `ai/settings.php`

Available Routing Actions:

- `get`
- `save`
- `test`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>csrf_token</code>	Field: csrf_token
<code>csrf_token</code>	Field: csrf_token

HTTP GET Query Parameters:

Parameter	Expected Content / Description
<code>action</code>	Routing control action

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

Endpoint File: bank-reconciliation.php**Available Routing Actions:**

- create_account
- add_transaction
- reconcile

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
account_name	Field: account_name
account_number	Unique reference number
bank_name	Field: bank_name
branch	Field: branch
currency	Field: currency
opening_balance	Field: opening_balance
gl_account_id	Field: gl_account_id
bank_account_id	Field: bank_account_id
transaction_date	Associated date timestamp
description	Detailed description or notes
reference	Field: reference
debit	Field: debit
credit	Field: credit
balance	Outstanding unpaid balance on the document
id	Primary key, unique identifier (auto-incrementing)

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/batches.php

POST

Endpoint File: batches.php

Available Routing Actions:

- create
- get_batches
- get_expiring

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
product_id	Reference to erp_products(id)
batch_number	Unique reference number
quantity	Line item quantity
expiry_date	Expiration date
cost_price	Standard purchase cost per unit
days	Field: days

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/crm-activities.php

POST/GET

Endpoint File: `crm-activities.php`

Available Routing Actions:

- `create`
- `complete`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>action</code>	Routing control action
<code>type</code>	Classification type
<code>subject</code>	Topic or header of the record
<code>description</code>	Detailed description or notes
<code>lead_id</code>	Reference to <code>erp_leads(id)</code>
<code>opportunity_id</code>	Reference to <code>erp_opportunities(id)</code>
<code>customer_id</code>	Reference to <code>erp_customers(id)</code>
<code>due_date</code>	Target completion date
<code>id</code>	Primary key, unique identifier (auto-incrementing)

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/customers.php

POST

Endpoint File: `customers.php`

Available Routing Actions:

- `create`
- `update`
- `delete`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>action</code>	Routing control action
<code>name</code>	Full name or name of the entity
<code>customer_code</code>	Unique alphanumeric business code for the customer
<code>email</code>	Email address
<code>phone</code>	Phone number
<code>address</code>	Street/mailling address
<code>city</code>	City
<code>country</code>	Country (default: Tanzania)
<code>tax_id</code>	TIN or VRN tax identification number
<code>credit_limit</code>	Maximum credit limit allowed for credit sales
<code>notes</code>	Internal remarks or customer-facing notes
<code>id</code>	Primary key, unique identifier (auto-incrementing)
<code>status</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/deliveries.php

POST

Endpoint File: deliveries.php

Available Routing Actions:

- create_from_so
- mark_delivered

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
order_id	Reference to erp_orders(id)
items	Field: items
customer_id	Reference to erp_customers(id)
delivery_date	Associated date timestamp
driver_name	Field: driver_name
vehicle_reg	Field: vehicle_reg
notes	Internal remarks or customer-facing notes
id	Primary key, unique identifier (auto-incrementing)

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```


`/api/delivery-notes.php`

POST

Endpoint File: `delivery-notes.php`

Available Routing Actions:

- `create`
- `update_status`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>action</code>	Routing control action
<code>customer_id</code>	Reference to <code>erp_customers(id)</code>
<code>date</code>	Document date
<code>items</code>	Field: <code>items</code>
<code>invoice_id</code>	Reference to <code>erp_invoices(id)</code>
<code>shipping_address</code>	Field: <code>shipping_address</code>
<code>driver_name</code>	Field: <code>driver_name</code>
<code>vehicle_number</code>	Unique reference number
<code>notes</code>	Internal remarks or customer-facing notes
<code>id</code>	Primary key, unique identifier (auto-incrementing)
<code>status</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/employees.php

POST

Endpoint File: `employees.php`

Available Routing Actions:

- `create`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>action</code>	Routing control action
<code>first_name</code>	Field: first_name
<code>last_name</code>	Field: last_name
<code>email</code>	Email address
<code>employee_code</code>	Unique identification code
<code>phone</code>	Phone number
<code>position</code>	Field: position
<code>join_date</code>	Associated date timestamp
<code>basic_salary</code>	Field: basic_salary
<code>status</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>bank_name</code>	Field: bank_name
<code>bank_account_number</code>	Unique reference number

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/expenses.php

POST

Endpoint File: expenses.php

Available Routing Actions:

- create

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
date	Document date
account_id	Reference to erp_accounts(id)
amount	Financial value or cost
payee	Field: payee
payment_method	Field: payment_method
description	Detailed description or notes
voucher_id	Field: voucher_id

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/get-participants.php

POST/GET

Endpoint File: `get-participants.php`

HTTP GET Query Parameters:

Parameter	Expected Content / Description
<code>meeting_id</code>	Reference to meeting room

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/get_notifications.php

POST/GET

Endpoint File: `get_notifications.php`

Available Routing Actions:

- `poll`
- `read`

HTTP GET Query Parameters:

Parameter	Expected Content / Description
<code>action</code>	Routing control action
<code>id</code>	Primary key, unique identifier (auto-incrementing)

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

`/api/get_user_email.php`

POST

Endpoint File: `get_user_email.php`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>full_name</code>	Full display name of the user

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/grn.php

POST

Endpoint File: grn.php

Available Routing Actions:

- create_from_po

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
po_id	Reference to erp_purchase_orders(id)
items	Field: items
supplier_id	Reference to erp_suppliers(id)
delivery_note_ref	Supplier delivery note reference number
notes	Internal remarks or customer-facing notes

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

`/api/invoices.php`

POST

Endpoint File: `invoices.php`

Available Routing Actions:

- `create`
- `update_status`
- `register_payment`
- `send_email`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>action</code>	Routing control action
<code>customer_id</code>	Reference to <code>erp_customers(id)</code>
<code>invoice_date</code>	Date the invoice was issued
<code>items</code>	Field: items
<code>invoice_number</code>	Unique system invoice number
<code>due_date</code>	Target completion date
<code>notes</code>	Internal remarks or customer-facing notes
<code>id</code>	Primary key, unique identifier (auto-incrementing)
<code>status</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/join-meeting.php

POST/GET

Endpoint File: join-meeting.php

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/journal-entries.php

POST

Endpoint File: journal-entries.php

Available Routing Actions:

- create

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
date	Document date
items	Field: items
description	Detailed description or notes

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```


Endpoint File: `leads.php`**Available Routing Actions:**

- `create`
- `update_status`
- `convert`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>action</code>	Routing control action
<code>first_name</code>	Field: first_name
<code>last_name</code>	Field: last_name
<code>email</code>	Email address
<code>phone</code>	Phone number
<code>company</code>	Field: company
<code>source</code>	Field: source
<code>assigned_to</code>	User ID of the assignee
<code>notes</code>	Internal remarks or customer-facing notes
<code>status</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
<code>id</code>	Primary key, unique identifier (auto-incrementing)

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/leave-meeting.php

POST/GET

Endpoint File: leave-meeting.php

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/leave.php

POST

Endpoint File: `leave.php`

Available Routing Actions:

- `create`
- `update_status`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>action</code>	Routing control action
<code>employee_id</code>	Reference to <code>erp_employees(id)</code>
<code>leave_type</code>	Type of leave (e.g., 'sick', 'annual')
<code>start_date</code>	Leave starting date
<code>end_date</code>	Leave ending date
<code>reason</code>	Explanation or justification
<code>id</code>	Primary key, unique identifier (auto-incrementing)
<code>status</code>	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/meeting-chat.php

POST/GET

Endpoint File: meeting-chat.php

Available Routing Actions:

- get_messages
- send_message

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
meeting_id	Reference to meeting room
message	Message text content

HTTP GET Query Parameters:

Parameter	Expected Content / Description
action	Routing control action
meeting_id	Reference to meeting room

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

`/api/opportunities.php`

POST/GET

Endpoint File: `opportunities.php`

Available Routing Actions:

- `create`
- `update_stage`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>action</code>	Routing control action
<code>name</code>	Full name or name of the entity
<code>customer_id</code>	Reference to <code>erp_customers(id)</code>
<code>lead_id</code>	Reference to <code>erp_leads(id)</code>
<code>amount</code>	Financial value or cost
<code>stage</code>	Sales pipeline stage
<code>probability</code>	Estimated win probability percentage
<code>expected_close_date</code>	Target close date of opportunity
<code>assigned_to</code>	User ID of the assignee
<code>id</code>	Primary key, unique identifier (auto-incrementing)

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/payroll.php

POST

Endpoint File: payroll.php

Available Routing Actions:

- run_payroll

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
month	Salary calculation month (YYYY-MM)
employees	List of employee IDs or JSON list

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/products.php

POST

Endpoint File: products.php

Available Routing Actions:

- create
- update
- delete

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
sku	Stock Keeping Unit (unique product code)
name	Full name or name of the entity
unit_price	Standard selling price per unit
description	Detailed description or notes
category_id	Reference to erp_categories(id)
supplier_id	Reference to erp_suppliers(id)
unit	Unit of measure (e.g., 'pcs', 'kg')
cost_price	Standard purchase cost per unit
stock_quantity	Current physical stock quantity on hand
reorder_level	Stock level threshold that triggers replenishment alerts
barcode	Universal product barcode number
id	Primary key, unique identifier (auto-incrementing)
status	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

Response Data Format (JSON):

```
{  
  "success": true,  
}
```

```
"message": "Operation completed successfully"
}
```

/api/purchase-orders.php

POST

Endpoint File: `purchase-orders.php`

Available Routing Actions:

- `create`
- `receive`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>action</code>	Routing control action
<code>supplier_id</code>	Reference to <code>erp_suppliers(id)</code>
<code>order_date</code>	Purchase order date
<code>items</code>	Field: items
<code>po_number</code>	Unique purchase order number
<code>expected_date</code>	Expected delivery date
<code>notes</code>	Internal remarks or customer-facing notes
<code>id</code>	Primary key, unique identifier (auto-incrementing)

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```


/api/purchase_requests.php

POST

Endpoint File: `purchase_requests.php`

Available Routing Actions:

- `create`
- `approve`
- `reject`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>action</code>	Routing control action
<code>items</code>	Field: items
<code>notes</code>	Internal remarks or customer-facing notes
<code>id</code>	Primary key, unique identifier (auto-incrementing)

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/quotes.php

POST

Endpoint File: quotes.php

Available Routing Actions:

- update_status
- create
- convert_to_invoice
- send_email

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
id	Primary key, unique identifier (auto-incrementing)
status	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')
customer_id	Reference to erp_customers(id)
date	Document date
items	Field: items
expiry_date	Expiration date
notes	Internal remarks or customer-facing notes

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/revenue_accounts.php

POST

Endpoint File: revenue_accounts.php

Available Routing Actions:

- list_main
- list_categories
- list_sub
- next_code
- create_main
- create_sub

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
parent_id	Self-referencing parent ID for hierarchical structures
name	Full name or name of the entity
description	Detailed description or notes

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/sales_orders.php

POST

Endpoint File: sales_orders.php

Available Routing Actions:

- create
- update_status
- convert_to_invoice

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
customer_id	Reference to erp_customers(id)
order_date	Purchase order date
items	Field: items
delivery_date	Associated date timestamp
notes	Internal remarks or customer-facing notes
id	Primary key, unique identifier (auto-incrementing)
status	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/save_user_email_settings.php

POST

Endpoint File: `save_user_email_settings.php`

HTTP POST Request Parameters:

Parameter	Expected Content / Description
<code>imap_host</code>	Field: imap_host
<code>imap_port</code>	Field: imap_port
<code>imap_user</code>	Field: imap_user
<code>imap_pass</code>	Field: imap_pass
<code>imap_ssl</code>	Field: imap_ssl
<code>smtp_host</code>	Field: smtp_host
<code>smtp_port</code>	Field: smtp_port
<code>smtp_user</code>	Field: smtp_user
<code>smtp_pass</code>	Field: smtp_pass
<code>smtp_ssl</code>	Field: smtp_ssl

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/settings.php

POST/GET

Endpoint File: settings.php

Available Routing Actions:

- update_company
- update_system
- update_numbering
- add_tax_rate
- update_role

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
company_name	Field: company_name
company_address	Field: company_address
company_phone	Field: company_phone
company_email	Field: company_email
company_vrn	Field: company_vrn
company_tin	Field: company_tin
currency	Field: currency
timezone	Field: timezone
date_format	Associated date timestamp
tax_enabled	Field: tax_enabled
default_tax_rate	Field: default_tax_rate
invoice_prefix	Field: invoice_prefix
invoice_next_number	Unique reference number
quote_prefix	Field: quote_prefix
quote_next_number	Unique reference number
po_prefix	Field: po_prefix
po_next_number	Unique reference number
name	Full name or name of the entity
rate	Field: rate
type	Classification type
is_default	Boolean flag for default settings (1 = Yes, 0 = No)
permissions	JSON/text representation of role permissions
role_id	Reference to erp_roles(id)

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/signal.php

POST/GET

Endpoint File: `signal.php`

Available Routing Actions:

- `send`
- `receive`
- `cleanup`

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```


/api/suppliers.php

POST

Endpoint File: suppliers.php

Available Routing Actions:

- create
- update
- delete

HTTP POST Request Parameters:

Parameter	Expected Content / Description
action	Routing control action
name	Full name or name of the entity
supplier_code	Unique alphanumeric business code for the supplier
contact_person	Field: contact_person
email	Email address
phone	Phone number
address	Street/mailling address
tax_id	TIN or VRN tax identification number
payment_terms	Standard payment terms (e.g., 'Net 30', 'COD')
id	Primary key, unique identifier (auto-incrementing)
status	Operational status or lifecycle state (e.g., 'active', 'pending', 'paid', 'posted')

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/test_user_email_connection.php

POST

Endpoint File: test_user_email_connection.php

HTTP POST Request Parameters:

Parameter	Expected Content / Description
imap_host	Field: imap_host
imap_port	Field: imap_port
imap_user	Field: imap_user
imap_pass	Field: imap_pass
imap_ssl	Field: imap_ssl
smtp_host	Field: smtp_host
smtp_port	Field: smtp_port
smtp_user	Field: smtp_user
smtp_pass	Field: smtp_pass
smtp_ssl	Field: smtp_ssl

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```

/api/update-video-status.php

POST/GET

Endpoint File: update-video-status.php

Response Data Format (JSON):

```
{
  "success": true,
  "message": "Operation completed successfully"
}
```